

135. Suppose that for each i such that $1 \leq i \leq N$ one has

$$\int_{i-1}^i f(t) dt = i^2. \text{ Show that } \int_0^N f(t) dt = \frac{N(N+1)(2N+1)}{6}.$$

138. If $\int_1^5 \sqrt{1+t^4} dt = 41.7133\dots$, what is $\int_1^5 \sqrt{1+u^4} du$?

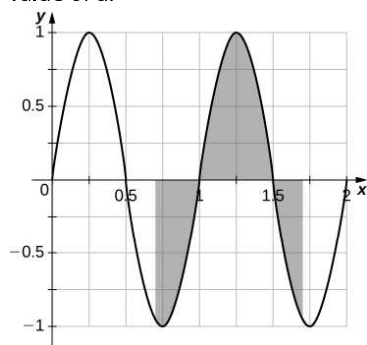
136. [T] Compute the left and right Riemann sums L_{10} and R_{10} and their average $\frac{L_{10}+R_{10}}{2}$ for $f(t) = t^2$ over $[0, 1]$. Given that $\int_0^1 t^2 dt = 0.\overline{33}$, to how many decimal places is $\frac{L_{10}+R_{10}}{2}$ accurate?

139. Estimate $\int_0^1 t dt$ using the left and right endpoint sums, each with a single rectangle. How does the average of these left and right endpoint sums compare with the actual value $\int_0^1 t dt$?

137. [T] Compute the left and right Riemann sums, L_{10} and R_{10} , and their average $\frac{L_{10}+R_{10}}{2}$ for $f(t) = (4-t^2)$ over $[1, 2]$. Given that $\int_1^2 (4-t^2) dt = 1.\overline{66}$, to how many decimal places is $\frac{L_{10}+R_{10}}{2}$ accurate?

140. Estimate $\int_0^1 t dt$ by comparison with the area of a single rectangle with height equal to the value of t at the midpoint $t = \frac{1}{2}$. How does this midpoint estimate compare with the actual value $\int_0^1 t dt$?

141. From the graph of $\sin(2\pi x)$ shown:
- Explain why $\int_0^1 \sin(2\pi t) dt = 0$.
 - Explain why, in general, $\int_a^{a+1} \sin(2\pi t) dt = 0$ for any value of a .



142. If f is 1-periodic ($f(t+1) = f(t)$), odd, and integrable over $[0, 1]$, is it always true that $\int_0^1 f(t) dt = 0$?
143. If f is 1-periodic and $\int_0^1 f(t) dt = A$, is it necessarily true that $\int_a^{1+a} f(t) dt = A$ for all A ?

5.3 The Fundamental Theorem of Calculus

Learning Objectives

- 5.3.1 Describe the meaning of the Mean Value Theorem for Integrals.
- 5.3.2 State the meaning of the Fundamental Theorem of Calculus, Part 1.
- 5.3.3 Use the Fundamental Theorem of Calculus, Part 1, to evaluate derivatives of integrals.
- 5.3.4 State the meaning of the Fundamental Theorem of Calculus, Part 2.
- 5.3.5 Use the Fundamental Theorem of Calculus, Part 2, to evaluate definite integrals.
- 5.3.6 Explain the relationship between differentiation and integration.

In the previous two sections, we looked at the definite integral and its relationship to the area under the curve of a function. Unfortunately, so far, the only tools we have available to calculate the value of a definite integral are geometric area formulas and limits of Riemann sums, and both approaches are extremely cumbersome. In this section we look at some more powerful and useful techniques for evaluating definite integrals.

These new techniques rely on the relationship between differentiation and integration. This relationship was discovered and explored by both Sir Isaac Newton and Gottfried Wilhelm Leibniz (among others) during the late 1600s and early

1700s, and it is codified in what we now call the **Fundamental Theorem of Calculus**, which has two parts that we examine in this section. Its very name indicates how central this theorem is to the entire development of calculus.

MEDIA

Isaac Newton's contributions to mathematics and physics changed the way we look at the world. The relationships he discovered, codified as Newton's laws and the law of universal gravitation, are still taught as foundational material in physics today, and his calculus has spawned entire fields of mathematics. To learn more, read a [brief biography](http://www.openstax.org/l/20_newtonbio) (http://www.openstax.org/l/20_newtonbio) of Newton with multimedia clips.

Before we get to this crucial theorem, however, let's examine another important theorem, the Mean Value Theorem for Integrals, which is needed to prove the Fundamental Theorem of Calculus.

The Mean Value Theorem for Integrals

The **Mean Value Theorem for Integrals** states that a continuous function on a closed interval takes on its average value at some point in that interval. The theorem guarantees that if $f(x)$ is continuous, a point c exists in an interval $[a, b]$ such that the value of the function at c is equal to the average value of $f(x)$ over $[a, b]$. We state this theorem mathematically with the help of the formula for the average value of a function that we presented at the end of the preceding section.

Theorem 5.3

The Mean Value Theorem for Integrals

If $f(x)$ is continuous over an interval $[a, b]$, then there is at least one point $c \in [a, b]$ such that

$$f(c) = \frac{1}{b-a} \int_a^b f(x) dx. \quad (5.15)$$

This formula can also be stated as

$$\int_a^b f(x) dx = f(c)(b-a).$$

Proof

Since $f(x)$ is continuous on $[a, b]$, by the extreme value theorem (see [Maxima and Minima](#)), it assumes minimum and maximum values— m and M , respectively—on $[a, b]$. Then, for all x in $[a, b]$, we have $m \leq f(x) \leq M$. Therefore, by the comparison theorem (see [The Definite Integral](#)), we have

$$m(b-a) \leq \int_a^b f(x) dx \leq M(b-a).$$

Dividing by $b-a$ gives us

$$m \leq \frac{1}{b-a} \int_a^b f(x) dx \leq M.$$

Since $\frac{1}{b-a} \int_a^b f(x) dx$ is a number between m and M , and since $f(x)$ is continuous and assumes the values m and M over $[a, b]$, by the Intermediate Value Theorem (see [Continuity](#)), there is a number c over $[a, b]$ such that

$$f(c) = \frac{1}{b-a} \int_a^b f(x) dx,$$

and the proof is complete.

□

EXAMPLE 5.15**Finding the Average Value of a Function**

Find the average value of the function $f(x) = 8 - 2x$ over the interval $[0, 4]$ and find c such that $f(c)$ equals the average value of the function over $[0, 4]$.

✓ Solution

The formula states the mean value of $f(x)$ is given by

$$\frac{1}{4-0} \int_0^4 (8-2x) dx.$$

We can see in [Figure 5.26](#) that the function represents a straight line and forms a right triangle bounded by the x - and y -axes. The area of the triangle is $A = \frac{1}{2}(\text{base})(\text{height})$. We have

$$A = \frac{1}{2}(4)(8) = 16.$$

The average value is found by multiplying the area by $1/(4-0)$. Thus, the average value of the function is

$$\frac{1}{4}(16) = 4.$$

Set the average value equal to $f(c)$ and solve for c .

$$\begin{aligned} 8 - 2c &= 4 \\ c &= 2 \end{aligned}$$

At $c = 2$, $f(2) = 4$.

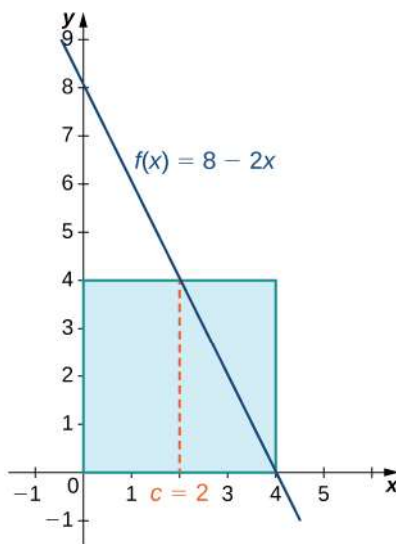


Figure 5.26 By the Mean Value Theorem, the continuous function $f(x)$ takes on its average value at c at least once over a closed interval.

- ✓ 5.14 Find the average value of the function $f(x) = \frac{x}{2}$ over the interval $[0, 6]$ and find c such that $f(c)$ equals the average value of the function over $[0, 6]$.

EXAMPLE 5.16**Finding the Point Where a Function Takes on Its Average Value**

Given $\int_0^3 x^2 dx = 9$, find c such that $f(c)$ equals the average value of $f(x) = x^2$ over $[0, 3]$.

☑ **Solution**

We are looking for the value of c such that

$$f(c) = \frac{1}{3-0} \int_0^3 x^2 dx = \frac{1}{3}(9) = 3.$$

Replacing $f(c)$ with c^2 , we have

$$\begin{aligned} c^2 &= 3 \\ c &= \pm\sqrt{3}. \end{aligned}$$

Since $-\sqrt{3}$ is outside the interval, take only the positive value. Thus, $c = \sqrt{3}$ (Figure 5.27).

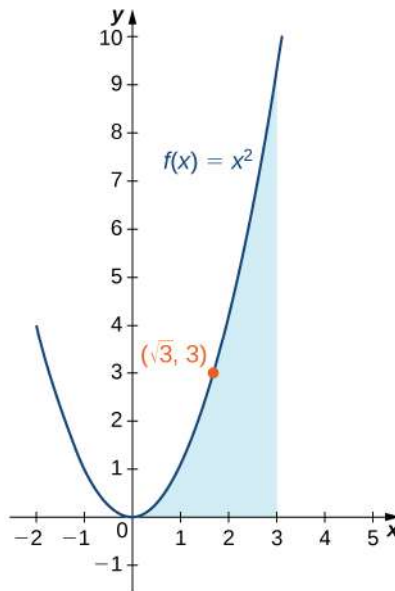


Figure 5.27 Over the interval $[0, 3]$, the function $f(x) = x^2$ takes on its average value at $c = \sqrt{3}$.

- ☑ 5.15 Given $\int_0^3 (2x^2 - 1) dx = 15$, find c such that $f(c)$ equals the average value of $f(x) = 2x^2 - 1$ over $[0, 3]$.

Fundamental Theorem of Calculus Part 1: Integrals and Antiderivatives

As mentioned earlier, the Fundamental Theorem of Calculus is an extremely powerful theorem that establishes the relationship between differentiation and integration, and gives us a way to evaluate definite integrals without using Riemann sums or calculating areas. The theorem is comprised of two parts, the first of which, the **Fundamental Theorem of Calculus, Part 1**, is stated here. Part 1 establishes the relationship between differentiation and integration.

Theorem 5.4

Fundamental Theorem of Calculus, Part 1

If $f(x)$ is continuous over an interval $[a, b]$, and the function $F(x)$ is defined by

$$F(x) = \int_a^x f(t) dt, \quad (5.16)$$

then $F'(x) = f(x)$ over (a, b) .

Before we delve into the proof, a couple of subtleties are worth mentioning here. First, a comment on the notation. Note

that we have defined a function, $F(x)$, as the definite integral of another function, $f(t)$, from the point a to the point x . At first glance, this is confusing, because we have said several times that a definite integral is a number, and here it looks like it's a function. The key here is to notice that for any particular value of x , the definite integral is a number. So the function $F(x)$ returns a number (the value of the definite integral) for each value of x .

Second, it is worth commenting on some of the key implications of this theorem. There is a reason it is called the *Fundamental Theorem of Calculus*. Not only does it establish a relationship between integration and differentiation, but also it guarantees that any integrable function has an antiderivative.

Proof

Applying the definition of the derivative, we have

$$\begin{aligned} F'(x) &= \lim_{h \rightarrow 0} \frac{F(x+h) - F(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{1}{h} \left[\int_a^{x+h} f(t) dt - \int_a^x f(t) dt \right] \\ &= \lim_{h \rightarrow 0} \frac{1}{h} \left[\int_a^{x+h} f(t) dt + \int_x^a f(t) dt \right] \\ &= \lim_{h \rightarrow 0} \frac{1}{h} \int_x^{x+h} f(t) dt. \end{aligned}$$

Looking carefully at this last expression, we see $\frac{1}{h} \int_x^{x+h} f(t) dt$ is just the average value of the function $f(x)$ over the interval $[x, x+h]$. Therefore, by [The Mean Value Theorem for Integrals](#), there is some number c in $[x, x+h]$ such that

$$\frac{1}{h} \int_x^{x+h} f(x) dx = f(c).$$

In addition, since c is between x and $x+h$, c approaches x as h approaches zero. Also, since $f(x)$ is continuous, we have $\lim_{h \rightarrow 0} f(c) = \lim_{c \rightarrow x} f(c) = f(x)$. Putting all these pieces together, we have

$$\begin{aligned} F'(x) &= \lim_{h \rightarrow 0} \frac{1}{h} \int_x^{x+h} f(x) dx \\ &= \lim_{h \rightarrow 0} f(c) \\ &= f(x), \end{aligned}$$

and the proof is complete.

□

EXAMPLE 5.17

Finding a Derivative with the Fundamental Theorem of Calculus

Use the [Fundamental Theorem of Calculus, Part 1](#) to find the derivative of

$$g(x) = \int_1^x \frac{1}{t^3 + 1} dt.$$

Solution

According to the Fundamental Theorem of Calculus, the derivative is given by

$$g'(x) = \frac{1}{x^3 + 1}.$$

- ✓ 5.16 Use the Fundamental Theorem of Calculus, Part 1 to find the derivative of $g(r) = \int_0^r \sqrt{x^2 + 4} dx$.

EXAMPLE 5.18**Using the Fundamental Theorem and the Chain Rule to Calculate Derivatives**

Let $F(x) = \int_1^{\sqrt{x}} \sin t dt$. Find $F'(x)$.

✓ **Solution**

Letting $u(x) = \sqrt{x}$, we have $F(x) = \int_1^{u(x)} \sin t dt$. Thus, by the Fundamental Theorem of Calculus and the chain rule,

$$\begin{aligned} F'(x) &= \sin(u(x)) \frac{du}{dx} \\ &= \sin(u(x)) \cdot \left(\frac{1}{2}x^{-1/2}\right) \\ &= \frac{\sin \sqrt{x}}{2\sqrt{x}}. \end{aligned}$$

- ✓ 5.17 Let $F(x) = \int_1^{x^3} \cos t dt$. Find $F'(x)$.

EXAMPLE 5.19**Using the Fundamental Theorem of Calculus with Two Variable Limits of Integration**

Let $F(x) = \int_x^{2x} t^3 dt$. Find $F'(x)$.

✓ **Solution**

We have $F(x) = \int_x^{2x} t^3 dt$. Both limits of integration are variable, so we need to split this into two integrals. We get

$$\begin{aligned} F(x) &= \int_x^{2x} t^3 dt \\ &= \int_x^0 t^3 dt + \int_0^{2x} t^3 dt \\ &= -\int_0^x t^3 dt + \int_0^{2x} t^3 dt. \end{aligned}$$

Differentiating the first term, we obtain

$$\frac{d}{dx} \left[-\int_0^x t^3 dt \right] = -x^3.$$

Differentiating the second term, we first let $u(x) = 2x$. Then,

$$\begin{aligned} \frac{d}{dx} \left[\int_0^{2x} t^3 dt \right] &= \frac{d}{dx} \left[\int_0^{u(x)} t^3 dt \right] \\ &= (u(x))^3 \frac{du}{dx} \\ &= (2x)^3 \cdot 2 \\ &= 16x^3. \end{aligned}$$

Thus,

$$\begin{aligned} F'(x) &= \frac{d}{dx} \left[-\int_0^x t^3 dt \right] + \frac{d}{dx} \left[\int_0^{2x} t^3 dt \right] \\ &= -x^3 + 16x^3 \\ &= 15x^3. \end{aligned}$$

✓ 5.18 Let $F(x) = \int_x^{x^2} \cos t dt$. Find $F'(x)$.

Fundamental Theorem of Calculus, Part 2: The Evaluation Theorem

The Fundamental Theorem of Calculus, Part 2, is perhaps the most important theorem in calculus. After tireless efforts by mathematicians for approximately 500 years, new techniques emerged that provided scientists with the necessary tools to explain many phenomena. Using calculus, astronomers could finally determine distances in space and map planetary orbits. Everyday financial problems such as calculating marginal costs or predicting total profit could now be handled with simplicity and accuracy. Engineers could calculate the bending strength of materials or the three-dimensional motion of objects. Our view of the world was forever changed with calculus.

After finding approximate areas by adding the areas of n rectangles, the application of this theorem is straightforward by comparison. It almost seems too simple that the area of an entire curved region can be calculated by just evaluating an antiderivative at the first and last endpoints of an interval.

Theorem 5.5

The Fundamental Theorem of Calculus, Part 2

If f is continuous over the interval $[a, b]$ and $F(x)$ is any antiderivative of $f(x)$, then

$$\int_a^b f(x) dx = F(b) - F(a). \quad (5.17)$$

We often see the notation $F(x) \Big|_a^b$ to denote the expression $F(b) - F(a)$. We use this vertical bar and associated limits a and b to indicate that we should evaluate the function $F(x)$ at the upper limit (in this case, b), and subtract the value of the function $F(x)$ evaluated at the lower limit (in this case, a).

The **Fundamental Theorem of Calculus, Part 2** (also known as the **evaluation theorem**) states that if we can find an antiderivative for the integrand, then we can evaluate the definite integral by evaluating the antiderivative at the endpoints of the interval and subtracting.

Proof

Let $P = \{x_i\}$, $i = 0, 1, \dots, n$ be a regular partition of $[a, b]$. Then, we can write

$$\begin{aligned} F(b) - F(a) &= F(x_n) - F(x_0) \\ &= [F(x_n) - F(x_{n-1})] + [F(x_{n-1}) - F(x_{n-2})] + \dots + [F(x_1) - F(x_0)] \\ &= \sum_{i=1}^n [F(x_i) - F(x_{i-1})]. \end{aligned}$$

Now, we know F is an antiderivative of f over $[a, b]$, so by the Mean Value Theorem (see [The Mean Value Theorem](#)) for $i = 0, 1, \dots, n$ we can find c_i in $[x_{i-1}, x_i]$ such that

$$F(x_i) - F(x_{i-1}) = F'(c_i)(x_i - x_{i-1}) = f(c_i) \Delta x.$$

Then, substituting into the previous equation, we have

$$F(b) - F(a) = \sum_{i=1}^n f(c_i) \Delta x.$$

Taking the limit of both sides as $n \rightarrow \infty$, we obtain

$$\begin{aligned} F(b) - F(a) &= \lim_{n \rightarrow \infty} \sum_{i=1}^n f(c_i) \Delta x \\ &= \int_a^b f(x) dx. \end{aligned}$$

□

EXAMPLE 5.20

Evaluating an Integral with the Fundamental Theorem of Calculus

Use [The Fundamental Theorem of Calculus, Part 2](#) to evaluate

$$\int_{-2}^2 (t^2 - 4) dt.$$

✓ Solution

Recall the power rule for [Antiderivatives](#):

$$\text{If } y = x^n, \int x^n dx = \frac{x^{n+1}}{n+1} + C.$$

Use this rule to find the antiderivative of the function and then apply the theorem. We have

$$\begin{aligned} \int_{-2}^2 (t^2 - 4) dt &= \left. \frac{t^3}{3} - 4t \right|_{-2}^2 \\ &= \left[\frac{(2)^3}{3} - 4(2) \right] - \left[\frac{(-2)^3}{3} - 4(-2) \right] \\ &= \left(\frac{8}{3} - 8 \right) - \left(-\frac{8}{3} + 8 \right) \\ &= \frac{8}{3} - 8 + \frac{8}{3} - 8 \\ &= \frac{16}{3} - 16 \\ &= -\frac{32}{3}. \end{aligned}$$

🔍 Analysis

Notice that we did not include the “+ C ” term when we wrote the antiderivative. The reason is that, according to the Fundamental Theorem of Calculus, Part 2, *any* antiderivative works. So, for convenience, we chose the antiderivative with $C = 0$. If we had chosen another antiderivative, the constant term would have canceled out. This always happens when evaluating a definite integral.

The region of the area we just calculated is depicted in [Figure 5.28](#). Note that the region between the curve and the x -axis is all below the x -axis. Area is always positive, but a definite integral can still produce a negative number (a net signed area). For example, if this were a profit function, a negative number indicates the company is operating at a loss over the given interval.

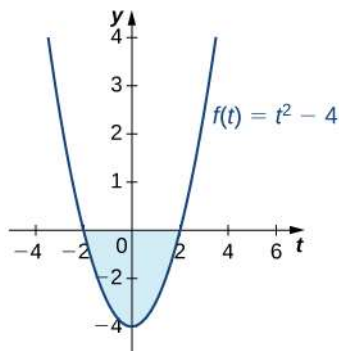


Figure 5.28 The evaluation of a definite integral can produce a negative value, even though area is always positive.

EXAMPLE 5.21

Evaluating a Definite Integral Using the Fundamental Theorem of Calculus, Part 2

Evaluate the following integral using the Fundamental Theorem of Calculus, Part 2:

$$\int_1^9 \frac{x-1}{\sqrt{x}} dx.$$

Solution

First, eliminate the radical by rewriting the integral using rational exponents. Then, separate the numerator terms by writing each one over the denominator:

$$\int_1^9 \frac{x-1}{x^{1/2}} dx = \int_1^9 \left(\frac{x}{x^{1/2}} - \frac{1}{x^{1/2}} \right) dx.$$

Use the properties of exponents to simplify:

$$\int_1^9 \left(\frac{x}{x^{1/2}} - \frac{1}{x^{1/2}} \right) dx = \int_1^9 (x^{1/2} - x^{-1/2}) dx.$$

Now, integrate using the power rule:

$$\begin{aligned} \int_1^9 (x^{1/2} - x^{-1/2}) dx &= \left(\frac{x^{3/2}}{\frac{3}{2}} - \frac{x^{1/2}}{\frac{1}{2}} \right) \bigg|_1^9 \\ &= \left[\frac{(9)^{3/2}}{\frac{3}{2}} - \frac{(9)^{1/2}}{\frac{1}{2}} \right] - \left[\frac{(1)^{3/2}}{\frac{3}{2}} - \frac{(1)^{1/2}}{\frac{1}{2}} \right] \\ &= \left[\frac{2}{3}(27) - 2(3) \right] - \left[\frac{2}{3}(1) - 2(1) \right] \\ &= 18 - 6 - \frac{2}{3} + 2 \\ &= \frac{40}{3}. \end{aligned}$$

See [Figure 5.29](#).

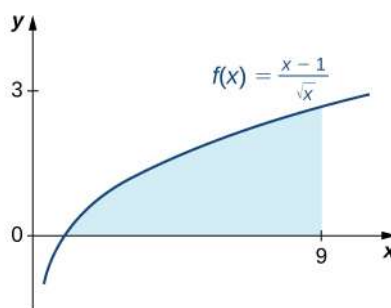


Figure 5.29 The area under the curve from $x = 1$ to $x = 9$ can be calculated by evaluating a definite integral.

- ✓ 5.19 Use [The Fundamental Theorem of Calculus, Part 2](#) to evaluate $\int_1^2 x^{-4} dx$.

EXAMPLE 5.22

A Roller-Skating Race

James and Kathy are racing on roller skates. They race along a long, straight track, and whoever has gone the farthest after 5 sec wins a prize. If James can skate at a velocity of $f(t) = 5 + 2t$ ft/sec and Kathy can skate at a velocity of $g(t) = 10 + \cos\left(\frac{\pi}{2}t\right)$ ft/sec, who is going to win the race?

✓ Solution

We need to integrate both functions over the interval $[0, 5]$ and see which value is bigger. For James, we want to calculate

$$\int_0^5 (5 + 2t) dt.$$

Using the power rule, we have

$$\begin{aligned} \int_0^5 (5 + 2t) dt &= (5t + t^2) \Big|_0^5 \\ &= (25 + 25) = 50. \end{aligned}$$

Thus, James has skated 50 ft after 5 sec. Turning now to Kathy, we want to calculate

$$\int_0^5 10 + \cos\left(\frac{\pi}{2}t\right) dt.$$

We know $\sin t$ is an antiderivative of $\cos t$, so it is reasonable to expect that an antiderivative of $\cos\left(\frac{\pi}{2}t\right)$ would involve $\sin\left(\frac{\pi}{2}t\right)$. However, when we differentiate $\sin\left(\frac{\pi}{2}t\right)$, we get $\frac{\pi}{2}\cos\left(\frac{\pi}{2}t\right)$ as a result of the chain rule, so we have to account for this additional coefficient when we integrate. We obtain

$$\begin{aligned} \int_0^5 10 + \cos\left(\frac{\pi}{2}t\right) dt &= \left(10t + \frac{2}{\pi}\sin\left(\frac{\pi}{2}t\right)\right) \Big|_0^5 \\ &= \left(50 + \frac{2}{\pi}\right) - \left(0 - \frac{2}{\pi}\sin 0\right) \\ &\approx 50.6. \end{aligned}$$

Kathy has skated approximately 50.6 ft after 5 sec. Kathy wins, but not by much!

- ✓ 5.20 Suppose James and Kathy have a rematch, but this time the official stops the contest after only 3 sec. Does this change the outcome?

STUDENT PROJECT

A Parachutist in Free Fall



Figure 5.30 Skydivers can adjust the velocity of their dive by changing the position of their body during the free fall. (credit: Jeremy T. Lock)

Julie is an avid skydiver. She has more than 300 jumps under her belt and has mastered the art of making adjustments to her body position in the air to control how fast she falls. If she arches her back and points her belly toward the ground, she reaches a terminal velocity of approximately 120 mph (176 ft/sec). If, instead, she orients her body with her head straight down, she falls faster, reaching a terminal velocity of 150 mph (220 ft/sec).

Since Julie will be moving (falling) in a downward direction, we assume the downward direction is positive to simplify our calculations. Julie executes her jumps from an altitude of 12,500 ft. After she exits the aircraft, she immediately starts falling at a velocity given by $v(t) = 32t$. She continues to accelerate according to this velocity function until she reaches terminal velocity. After she reaches terminal velocity, her speed remains constant until she pulls her ripcord and slows down to land.

On her first jump of the day, Julie orients herself in the slower “belly down” position (terminal velocity is 176 ft/sec). Using this information, answer the following questions.

1. How long after she exits the aircraft does Julie reach terminal velocity?
2. Based on your answer to question 1, set up an expression involving one or more integrals that represents the distance Julie falls after 30 sec.
3. If Julie pulls her ripcord at an altitude of 3000 ft, how long does she spend in a free fall?
4. Julie pulls her ripcord at 3000 ft. It takes 5 sec for her parachute to open completely and for her to slow down, during which time she falls another 400 ft. After her canopy is fully open, her speed is reduced to 16 ft/sec. Find the total time Julie spends in the air, from the time she leaves the airplane until the time her feet touch the ground.

On Julie's second jump of the day, she decides she wants to fall a little faster and orients herself in the “head down” position. Her terminal velocity in this position is 220 ft/sec. Answer these questions based on this velocity:

5. How long does it take Julie to reach terminal velocity in this case?
6. Before pulling her ripcord, Julie reorients her body in the “belly down” position so she is not moving quite as fast

when her parachute opens. If she begins this maneuver at an altitude of 4000 ft, how long does she spend in a free fall before beginning the reorientation?

Some jumpers wear “wingsuits” (see [Figure 5.31](#)). These suits have fabric panels between the arms and legs and allow the wearer to glide around in a free fall, much like a flying squirrel. (Indeed, the suits are sometimes called “flying squirrel suits.”) When wearing these suits, terminal velocity can be reduced to about 30 mph (44 ft/sec), allowing the wearers a much longer time in the air. Wingsuit flyers still use parachutes to land; although the vertical velocities are within the margin of safety, horizontal velocities can exceed 70 mph, much too fast to land safely.



Figure 5.31 The fabric panels on the arms and legs of a wingsuit work to reduce the vertical velocity of a skydiver's fall. (credit: Richard Schneider)

Answer the following question based on the velocity in a wingsuit.

7. If Julie dons a wingsuit before her third jump of the day, and she pulls her ripcord at an altitude of 3000 ft, how long does she get to spend gliding around in the air?



SECTION 5.3 EXERCISES

144. Consider two athletes running at variable speeds $v_1(t)$ and $v_2(t)$. The runners start and finish a race at exactly the same time. Explain why the two runners must be going the same speed at some point.
145. Two mountain climbers start their climb at base camp, taking two different routes, one steeper than the other, and arrive at the peak at exactly the same time. Is it necessarily true that, at some point, both climbers increased in altitude at the same rate?
146. To get on a certain toll road a driver has to take a card that lists the mile entrance point. The card also has a timestamp. When going to pay the toll at the exit, the driver is surprised to receive a speeding ticket along with the toll. Explain how this can happen.
147. Set $F(x) = \int_1^x (1-t) dt$. Find $F'(2)$ and the average value of F' over $[1, 2]$.

In the following exercises, use the Fundamental Theorem of Calculus, Part 1, to find each derivative.

148. $\frac{d}{dx} \int_1^x e^{-t^2} dt$

149. $\frac{d}{dx} \int_1^x e^{\cos t} dt$

150. $\frac{d}{dx} \int_3^x \sqrt{9-y^2} dy$

151. $\frac{d}{dx} \int_3^x \frac{ds}{\sqrt{16-s^2}}$

152. $\frac{d}{dx} \int_x^{2x} t dt$

153. $\frac{d}{dx} \int_0^{\sqrt{x}} t dt$

154. $\frac{d}{dx} \int_0^{\sin x} \sqrt{1-t^2} dt$

155. $\frac{d}{dx} \int_{\cos x}^1 \sqrt{1-t^2} dt$

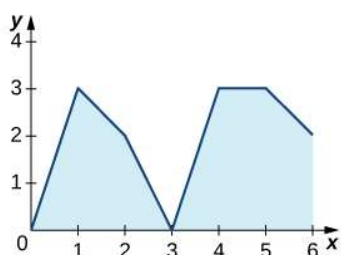
156. $\frac{d}{dx} \int_1^{\sqrt{x}} \frac{t^2}{1+t^4} dt$

157. $\frac{d}{dx} \int_1^{x^2} \frac{\sqrt{t}}{1+t} dt$

158. $\frac{d}{dx} \int_0^{\ln x} e^t dt$

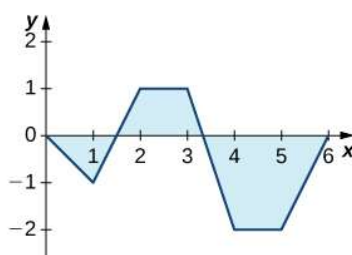
159. $\frac{d}{dx} \int_1^{e^x} \ln u^2 du$

160. The graph of $y = \int_0^x f(t) dt$, where f is a piecewise constant function, is shown here.



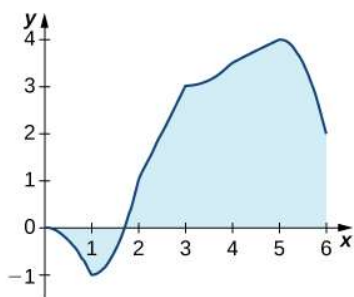
- Over which intervals is f positive? Over which intervals is it negative? Over which intervals, if any, is it equal to zero?
- What are the maximum and minimum values of f ?
- What is the average value of f ?

161. The graph of $y = \int_0^x f(t) dt$, where f is a piecewise constant function, is shown here.



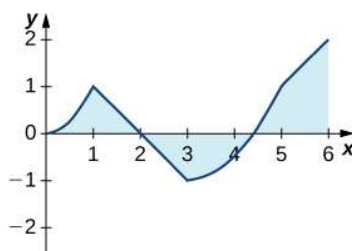
- Over which intervals is f positive? Over which intervals is it negative? Over which intervals, if any, is it equal to zero?
- What are the maximum and minimum values of f ?
- What is the average value of f ?

162. The graph of $y = \int_0^x \ell(t) dt$, where ℓ is a piecewise linear function, is shown here.



- Over which intervals is ℓ positive? Over which intervals is it negative? Over which, if any, is it zero?
- Over which intervals is ℓ increasing? Over which is it decreasing? Over which, if any, is it constant?
- What is the average value of ℓ ?

163. The graph of $y = \int_0^x \ell(t) dt$, where ℓ is a piecewise linear function, is shown here.



- Over which intervals is ℓ positive? Over which intervals is it negative? Over which, if any, is it zero?
- Over which intervals is ℓ increasing? Over which is it decreasing? Over which intervals, if any, is it constant?
- What is the average value of ℓ ?

In the following exercises, use a calculator to estimate the area under the curve by computing T_{10} , the average of the left- and right-endpoint Riemann sums using $N = 10$ rectangles. Then, using the Fundamental Theorem of Calculus, Part 2, determine the exact area.

164. [T] $y = x^2$ over $[0, 4]$ 165. [T] $y = x^3 + 6x^2 + x - 5$ over $[-4, 2]$ 166. [T] $y = \sqrt{x^3}$ over $[0, 6]$
167. [T] $y = \sqrt{x} + x^2$ over $[1, 9]$ 168. [T] $\int (\cos x - \sin x) dx$ over $[0, \pi]$ 169. [T] $\int \frac{4}{x^2} dx$ over $[1, 4]$

In the following exercises, evaluate each definite integral using the Fundamental Theorem of Calculus, Part 2.

170. $\int_{-1}^2 (x^2 - 3x) dx$ 171. $\int_{-2}^3 (x^2 + 3x - 5) dx$ 172. $\int_{-2}^3 (t + 2)(t - 3) dt$
173. $\int_2^3 (t^2 - 9)(4 - t^2) dt$ 174. $\int_1^2 x^9 dx$ 175. $\int_0^1 x^{99} dx$
176. $\int_4^8 (4t^{5/2} - 3t^{3/2}) dt$ 177. $\int_{1/4}^4 \left(x^2 - \frac{1}{x^2}\right) dx$ 178. $\int_1^2 \frac{2}{x^3} dx$
179. $\int_1^4 \frac{1}{2\sqrt{x}} dx$ 180. $\int_1^4 \frac{2 - \sqrt{t}}{t^2} dt$ 181. $\int_1^{16} \frac{dt}{t^{1/4}}$
182. $\int_0^{2\pi} \cos \theta d\theta$ 183. $\int_0^{\pi/2} \sin \theta d\theta$ 184. $\int_0^{\pi/4} \sec^2 \theta d\theta$
185. $\int_0^{\pi/4} \sec \theta \tan \theta d\theta$ 186. $\int_{\pi/3}^{\pi/4} \csc \theta \cot \theta d\theta$ 187. $\int_{\pi/4}^{\pi/2} \csc^2 \theta d\theta$
188. $\int_1^2 \left(\frac{1}{t^2} - \frac{1}{t^3}\right) dt$ 189. $\int_{-2}^{-1} \left(\frac{1}{t^2} - \frac{1}{t^3}\right) dt$

In the following exercises, use the evaluation theorem to express the integral as a function $F(x)$.

190. $\int_a^x t^2 dt$ 191. $\int_1^x e^t dt$ 192. $\int_0^x \cos t dt$
193. $\int_{-x}^x \sin t dt$

In the following exercises, identify the roots of the integrand to remove absolute values, then evaluate using the Fundamental Theorem of Calculus, Part 2.

194. $\int_{-2}^3 |x| dx$ 195. $\int_{-2}^4 |t^2 - 2t - 3| dt$ 196. $\int_0^\pi |\cos t| dt$

197. $\int_{-\pi/2}^{\pi/2} |\sin t| dt$

198. Suppose that the number of hours of daylight on a given day in Seattle is modeled by the function $-3.75 \cos\left(\frac{\pi t}{6}\right) + 12.25$, with t given in months and $t = 0$ corresponding to the winter solstice.
- What is the average number of daylight hours in a year?
 - At which times t_1 and t_2 , where $0 \leq t_1 < t_2 < 12$, do the number of daylight hours equal the average number?
 - Write an integral that expresses the total number of daylight hours in Seattle between t_1 and t_2 .
 - Compute the mean hours of daylight in Seattle between t_1 and t_2 , where $0 \leq t_1 < t_2 < 12$, and then between t_2 and t_1 , and show that the average of the two is equal to the average day length.
199. Suppose the rate of gasoline consumption over the course of a year in the United States can be modeled by a sinusoidal function of the form $\left(11.21 - \cos\left(\frac{\pi t}{6}\right)\right) \times 10^9$ gal/mo.
- What is the average monthly consumption, and for which values of t is the rate at time t equal to the average rate?
 - What is the number of gallons of gasoline consumed in the United States in a year?
 - Write an integral that expresses the average monthly U.S. gas consumption during the part of the year between the beginning of April ($t = 3$) and the end of September ($t = 9$).

200. Explain why, if f is continuous over $[a, b]$, there is at least one point $c \in [a, b]$ such that

$$f(c) = \frac{1}{b-a} \int_a^b f(t) dt.$$

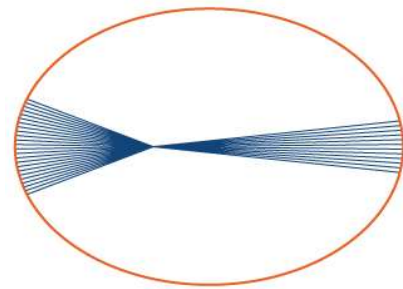
201. Explain why, if f is continuous over $[a, b]$ and is not equal to a constant, there is at least one point $M \in [a, b]$ such that

$$f(M) > \frac{1}{b-a} \int_a^b f(t) dt$$

and at least one point $m \in [a, b]$ such that

$$f(m) < \frac{1}{b-a} \int_a^b f(t) dt.$$

202. Kepler's first law states that the planets move in elliptical orbits with the Sun at one focus. The closest point of a planetary orbit to the Sun is called the *perihelion* (for Earth, it currently occurs around January 3) and the farthest point is called the *aphelion* (for Earth, it currently occurs around July 4). Kepler's second law states that planets sweep out equal areas of their elliptical orbits in equal times. Thus, the two arcs indicated in the following figure are swept out in equal times. At what time of year is Earth moving fastest in its orbit? When is it moving slowest?



203. A point on an ellipse with major axis length $2a$ and minor axis length $2b$ has the coordinates $(a \cos \theta, b \sin \theta)$, $0 \leq \theta \leq 2\pi$.
- Show that the distance from this point to the focus at $(-c, 0)$ is $d(\theta) = a + c \cos \theta$, where $c = \sqrt{a^2 - b^2}$.
 - Use these coordinates to show that the average distance $\bar{d}$ from a point on the ellipse to the focus at $(-c, 0)$, with respect to angle θ , is a .

204. As implied earlier, according to Kepler's laws, Earth's orbit is an ellipse with the Sun at one focus. The perihelion for Earth's orbit around the Sun is 147,098,290 km and the aphelion is 152,098,232 km.
- By placing the major axis along the x -axis, find the average distance from Earth to the Sun.
 - The classic definition of an astronomical unit (AU) is the distance from Earth to the Sun, and its value was computed as the average of the perihelion and aphelion distances. Is this definition justified?

205. The force of gravitational attraction between the Sun and a planet is $F(\theta) = \frac{GmM}{r^2(\theta)}$, where m is the mass of the planet, M is the mass of the Sun, G is a universal constant, and $r(\theta)$ is the distance between the Sun and the planet when the planet is at an angle θ with the major axis of its orbit. Assuming that M , m , and the ellipse parameters a and b (half-lengths of the major and minor axes) are given, set up—but do not evaluate—an integral that expresses in terms of G , m , M , a , b the average gravitational force between the Sun and the planet.

- 206.** The displacement from rest of a mass attached to a spring satisfies the simple harmonic motion equation $x(t) = A \cos(\omega t - \phi)$, where ϕ is a phase constant, ω is the angular frequency, and A is the amplitude. Find the average velocity, the average speed (magnitude of velocity), the average displacement, and the average distance from rest (magnitude of displacement) of the mass.

5.4 Integration Formulas and the Net Change Theorem

Learning Objectives

- 5.4.1** Apply the basic integration formulas.
- 5.4.2** Explain the significance of the net change theorem.
- 5.4.3** Use the net change theorem to solve applied problems.
- 5.4.4** Apply the integrals of odd and even functions.

In this section, we use some basic integration formulas studied previously to solve some key applied problems. It is important to note that these formulas are presented in terms of *indefinite* integrals. Although definite and indefinite integrals are closely related, there are some key differences to keep in mind. A definite integral is either a number (when the limits of integration are constants) or a single function (when one or both of the limits of integration are variables). An indefinite integral represents a family of functions, all of which differ by a constant. As you become more familiar with integration, you will get a feel for when to use definite integrals and when to use indefinite integrals. You will naturally select the correct approach for a given problem without thinking too much about it. However, until these concepts are cemented in your mind, think carefully about whether you need a definite integral or an indefinite integral and make sure you are using the proper notation based on your choice.

Basic Integration Formulas

Recall the integration formulas given in the [table in Antiderivatives](#) and the rule on properties of definite integrals. Let's look at a few examples of how to apply these rules.

EXAMPLE 5.23

Integrating a Function Using the Power Rule

Use the power rule to integrate the function $\int_1^4 \sqrt{t}(1+t) dt$.

Solution

The first step is to rewrite the function and simplify it so we can apply the power rule:

$$\begin{aligned} \int_1^4 \sqrt{t}(1+t) dt &= \int_1^4 t^{1/2}(1+t) dt \\ &= \int_1^4 (t^{1/2} + t^{3/2}) dt. \end{aligned}$$

Now apply the power rule:

$$\begin{aligned}
 \int_1^4 (t^{1/2} + t^{3/2}) dt &= \left(\frac{2}{3} t^{3/2} + \frac{2}{5} t^{5/2} \right) \Big|_1^4 \\
 &= \left[\frac{2}{3} (4)^{3/2} + \frac{2}{5} (4)^{5/2} \right] - \left[\frac{2}{3} (1)^{3/2} + \frac{2}{5} (1)^{5/2} \right] \\
 &= \frac{256}{15}.
 \end{aligned}$$

- ✓ 5.21 Find the definite integral of $f(x) = x^2 - 3x$ over the interval $[1, 3]$.

The Net Change Theorem

The **net change theorem** considers the integral of a *rate of change*. It says that when a quantity changes, the new value equals the initial value plus the integral of the rate of change of that quantity. The formula can be expressed in two ways. The second is more familiar; it is simply the definite integral.

Theorem 5.6

Net Change Theorem

The new value of a changing quantity equals the initial value plus the integral of the rate of change:

$$\begin{aligned}
 F(b) &= F(a) + \int_a^b F'(x) dx \\
 &\text{or} \\
 \int_a^b F'(x) dx &= F(b) - F(a).
 \end{aligned} \tag{5.18}$$

Subtracting $F(a)$ from both sides of the first equation yields the second equation. Since they are equivalent formulas, which one we use depends on the application.

The significance of the net change theorem lies in the results. Net change can be applied to area, distance, and volume, to name only a few applications. Net change accounts for negative quantities automatically without having to write more than one integral. To illustrate, let's apply the net change theorem to a velocity function in which the result is displacement.

We looked at a simple example of this in [The Definite Integral](#). Suppose a car is moving due north (the positive direction) at 40 mph between 2 p.m. and 4 p.m., then the car moves south at 30 mph between 4 p.m. and 5 p.m. We can graph this motion as shown in [Figure 5.32](#).

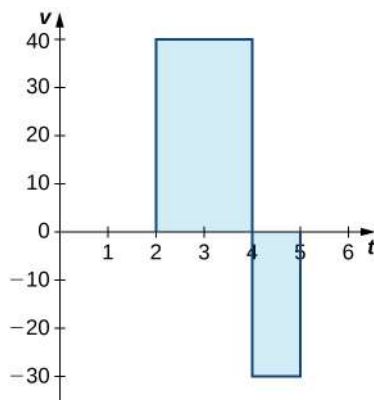


Figure 5.32 The graph shows speed versus time for the given motion of a car.

Just as we did before, we can use definite integrals to calculate the net displacement as well as the total distance traveled. The net displacement is given by

$$\begin{aligned}
 \int_2^5 v(t) dt &= \int_2^4 40 dt + \int_4^5 -30 dt \\
 &= 80 - 30 \\
 &= 50.
 \end{aligned}$$

Thus, at 5 p.m. the car is 50 mi north of its starting position. The total distance traveled is given by

$$\begin{aligned}
 \int_2^5 |v(t)| dt &= \int_2^4 40 dt + \int_4^5 30 dt \\
 &= 80 + 30 \\
 &= 110.
 \end{aligned}$$

Therefore, between 2 p.m. and 5 p.m., the car traveled a total of 110 mi.

To summarize, net displacement may include both positive and negative values. In other words, the velocity function accounts for both forward distance and backward distance. To find net displacement, integrate the velocity function over the interval. Total distance traveled, on the other hand, is always positive. To find the total distance traveled by an object, regardless of direction, we need to integrate the absolute value of the velocity function.

EXAMPLE 5.24

Finding Net Displacement

Given a velocity function $v(t) = 3t - 5$ (in meters per second) for a particle in motion from time $t = 0$ to time $t = 3$, find the net displacement of the particle.

Solution

Applying the net change theorem, we have

$$\begin{aligned}
 \int_0^3 (3t - 5) dt &= \left. \frac{3t^2}{2} - 5t \right|_0^3 \\
 &= \left[\frac{3(3)^2}{2} - 5(3) \right] - 0 \\
 &= \frac{27}{2} - 15 \\
 &= \frac{27}{2} - \frac{30}{2} \\
 &= -\frac{3}{2}.
 \end{aligned}$$

The net displacement is $-\frac{3}{2}$ m ([Figure 5.33](#)).

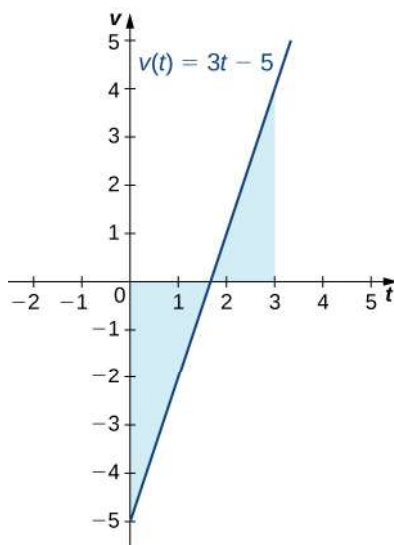


Figure 5.33 The graph shows velocity versus time for a particle moving with a linear velocity function.

EXAMPLE 5.25

Finding the Total Distance Traveled

Use [Example 5.24](#) to find the total distance traveled by a particle according to the velocity function $v(t) = 3t - 5$ m/sec over a time interval $[0, 3]$.

✓ Solution

The total distance traveled includes both the positive and the negative values. Therefore, we must integrate the absolute value of the velocity function to find the total distance traveled.

To continue with the example, use two integrals to find the total distance. First, find the t -intercept of the function, since that is where the division of the interval occurs. Set the equation equal to zero and solve for t . Thus,

$$\begin{aligned} 3t - 5 &= 0 \\ 3t &= 5 \\ t &= \frac{5}{3}. \end{aligned}$$

The two subintervals are $[0, \frac{5}{3}]$ and $[\frac{5}{3}, 3]$. To find the total distance traveled, integrate the absolute value of the function. Since the function is negative over the interval $[0, \frac{5}{3}]$, we have $|v(t)| = -v(t)$ over that interval. Over $[\frac{5}{3}, 3]$, the function is positive, so $|v(t)| = v(t)$. Thus, we have

$$\begin{aligned} \int_0^3 |v(t)| dt &= \int_0^{5/3} -v(t) dt + \int_{5/3}^3 v(t) dt \\ &= \int_0^{5/3} 5 - 3t dt + \int_{5/3}^3 3t - 5 dt \\ &= \left(5t - \frac{3t^2}{2} \right) \Big|_0^{5/3} + \left(\frac{3t^2}{2} - 5t \right) \Big|_{5/3}^3 \\ &= \left[5 \left(\frac{5}{3} \right) - \frac{3(5/3)^2}{2} \right] - 0 + \left[\frac{27}{2} - 15 \right] - \left[\frac{3(5/3)^2}{2} - \frac{25}{3} \right] \\ &= \frac{25}{3} - \frac{25}{6} + \frac{27}{2} - 15 - \frac{25}{6} + \frac{25}{3} \\ &= \frac{41}{6}. \end{aligned}$$

So, the total distance traveled is $\frac{41}{6}$ m.

- ✓ 5.22 Find the net displacement and total distance traveled in meters given the velocity function $f(t) = \frac{1}{2}e^t - 2$ over the interval $[0, 2]$.

Applying the Net Change Theorem

The net change theorem can be applied to the flow and consumption of fluids, as shown in [Example 5.26](#).

EXAMPLE 5.26

How Many Gallons of Gasoline Are Consumed?

If the motor on a motorboat is started at $t = 0$ and the boat consumes gasoline at the rate of $5 - 0.1t^3$ gal/hr, how much gasoline is used in the first 2 hours?

Solution

Express the problem as a definite integral, integrate, and evaluate using the Fundamental Theorem of Calculus. The limits of integration are the endpoints of the interval $[0, 2]$. We have

$$\begin{aligned}\int_0^2 (5 - 0.1t^3) dt &= \left(5t - 0.1 \frac{t^4}{4} \right) \bigg|_0^2 \\ &= \left[5 \left(2 \right) - 0.1 \frac{(2)^4}{4} \right] - 0 \\ &= 10 - 0.4 \\ &= 9.6\end{aligned}$$

Thus, the motorboat uses 9.6 gal of gas in 2 hours.

EXAMPLE 5.27

Chapter Opener: Iceboats



Figure 5.34 (credit: modification of work by Carter Brown, Flickr)

As we saw at the beginning of the chapter, top iceboat racers ([Figure 5.1](#)) can attain speeds of up to five times the wind speed. Andrew is an intermediate iceboater, though, so he attains speeds equal to only twice the wind speed. Suppose Andrew takes his iceboat out one morning when a light 5-mph breeze has been blowing all morning. As Andrew gets his iceboat set up, though, the wind begins to pick up. During his first half hour of iceboating, the wind speed increases according to the function $v(t) = 20t + 5$. For the second half hour of Andrew's outing, the wind remains steady at 15

mph. In other words, the wind speed is given by

$$v(t) = \begin{cases} 20t + 5 & \text{for } 0 \leq t \leq \frac{1}{2} \\ 15 & \text{for } \frac{1}{2} \leq t \leq 1. \end{cases}$$

Recalling that Andrew's iceboat travels at twice the wind speed, and assuming he moves in a straight line away from his starting point, how far is Andrew from his starting point after 1 hour?

 Solution

To figure out how far Andrew has traveled, we need to integrate his velocity, which is twice the wind speed. Then

$$\text{Distance} = \int_0^1 2v(t) dt.$$

Substituting the expressions we were given for $v(t)$, we get

$$\begin{aligned} \int_0^1 2v(t) dt &= \int_0^{1/2} 2v(t) dt + \int_{1/2}^1 2v(t) dt \\ &= \int_0^{1/2} 2(20t + 5) dt + \int_{1/2}^1 2(15) dt \\ &= \int_0^{1/2} (40t + 10) dt + \int_{1/2}^1 30 dt \\ &= [20t^2 + 10t]_0^{1/2} + [30t]_{1/2}^1 \\ &= \left(\frac{20}{4} + 5\right) - 0 + (30 - 15) \\ &= 25. \end{aligned}$$

Andrew is 25 mi from his starting point after 1 hour.

-  5.23 Suppose that, instead of remaining steady during the second half hour of Andrew's outing, the wind starts to die down according to the function $v(t) = -10t + 20$. In other words, the wind speed is given by

$$v(t) = \begin{cases} 20t + 5 & \text{for } 0 \leq t \leq \frac{1}{2} \\ -10t + 15 & \text{for } \frac{1}{2} \leq t \leq 1. \end{cases}$$

Under these conditions, how far from his starting point is Andrew after 1 hour?

Integrating Even and Odd Functions

We saw in [Functions and Graphs](#) that an even function is a function in which $f(-x) = f(x)$ for all x in the domain—that is, the graph of the curve is unchanged when x is replaced with $-x$. The graphs of even functions are symmetric about the y -axis. An odd function is one in which $f(-x) = -f(x)$ for all x in the domain, and the graph of the function is symmetric about the origin.

Integrals of even functions, when the limits of integration are from $-a$ to a , involve two equal areas, because they are symmetric about the y -axis. Integrals of odd functions, when the limits of integration are similarly $[-a, a]$, evaluate to zero because the areas above and below the x -axis are equal.

Rule: Integrals of Even and Odd Functions

For continuous even functions such that $f(-x) = f(x)$,

$$\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx.$$

For continuous odd functions such that $f(-x) = -f(x)$,

$$\int_{-a}^a f(x) dx = 0.$$

EXAMPLE 5.28**Integrating an Even Function**

Integrate the even function $\int_{-2}^2 (3x^8 - 2) dx$ and verify that the integration formula for even functions holds.

✓ Solution

The symmetry appears in the graphs in [Figure 5.35](#). Graph (a) shows the region below the curve and above the x-axis. We have to zoom in to this graph by a huge amount to see the region. Graph (b) shows the region above the curve and below the x-axis. The signed area of this region is negative. Both views illustrate the symmetry about the y-axis of an even function. We have

$$\begin{aligned} \int_{-2}^2 (3x^8 - 2) dx &= \left(\frac{x^9}{3} - 2x \right) \Big|_{-2}^2 \\ &= \left[\frac{(2)^9}{3} - 2(2) \right] - \left[\frac{(-2)^9}{3} - 2(-2) \right] \\ &= \left(\frac{512}{3} - 4 \right) - \left(-\frac{512}{3} + 4 \right) \\ &= \frac{1000}{3}. \end{aligned}$$

To verify the integration formula for even functions, we can calculate the integral from 0 to 2 and double it, then check to make sure we get the same answer.

$$\begin{aligned} \int_0^2 (3x^8 - 2) dx &= \left(\frac{x^9}{3} - 2x \right) \Big|_0^2 \\ &= \frac{512}{3} - 4 \\ &= \frac{500}{3} \end{aligned}$$

Since $2 \cdot \frac{500}{3} = \frac{1000}{3}$, we have verified the formula for even functions in this particular example.

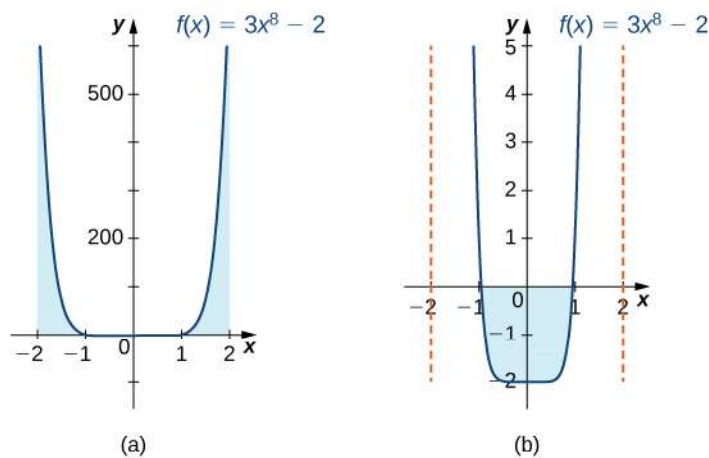


Figure 5.35 Graph (a) shows the positive area between the curve and the x-axis, whereas graph (b) shows the negative area between the curve and the x-axis. Both views show the symmetry about the y-axis.

EXAMPLE 5.29**Integrating an Odd Function**

Evaluate the definite integral of the odd function $-5 \sin x$ over the interval $[-\pi, \pi]$.

✓ Solution

The graph is shown in [Figure 5.36](#). We can see the symmetry about the origin by the positive area above the x -axis over $[-\pi, 0]$, and the negative area below the x -axis over $[0, \pi]$. We have

$$\begin{aligned} \int_{-\pi}^{\pi} -5 \sin x \, dx &= -5(-\cos x) \Big|_{-\pi}^{\pi} \\ &= 5 \cos x \Big|_{-\pi}^{\pi} \\ &= [5 \cos \pi] - [5 \cos(-\pi)] \\ &= -5 - (-5) \\ &= 0. \end{aligned}$$

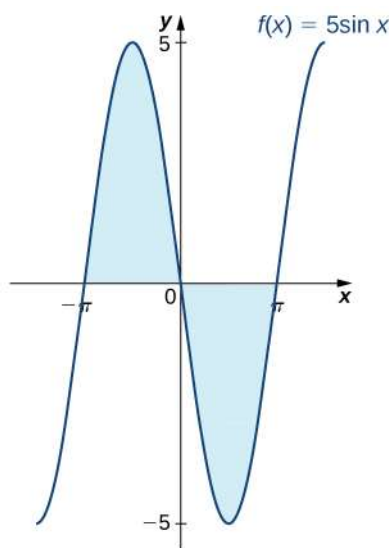


Figure 5.36 The graph shows areas between a curve and the x -axis for an odd function.

✓ 5.24 Integrate the function $\int_{-2}^2 x^4 \, dx$.

**SECTION 5.4 EXERCISES**

Use basic integration formulas to compute the following antiderivatives or definite integrals.

207. $\int \left(\sqrt{x} - \frac{1}{\sqrt{x}} \right) dx$

208. $\int \left(e^{2x} - \frac{1}{2} e^{x/2} \right) dx$

209. $\int \frac{dx}{2x}$

210. $\int \frac{x-1}{x^2} dx$

211. $\int_0^{\pi} (\sin x - \cos x) dx$

212. $\int_0^{\pi/2} (x - \sin x) dx$

- 213.** Write an integral that expresses the increase in the perimeter $P(s)$ of a square when its side length s increases from 2 units to 4 units and evaluate the integral.
- 214.** Write an integral that quantifies the change in the area $A(s) = s^2$ of a square when the side length doubles from 5 units to 25 units and evaluate the integral.
- 215.** A regular N -gon (an N -sided polygon with sides that have equal length s , such as a pentagon or hexagon) has perimeter Ns . Write an integral that expresses the increase in perimeter of a regular N -gon when the length of each side increases from 1 unit to 2 units and evaluate the integral.
- 216.** The area of a regular pentagon with side length $a > 0$ is pa^2 with $p = \frac{1}{4}\sqrt{5(5 + 2\sqrt{5})}$. The Pentagon in Washington, DC, has inner sides of length 360 ft and outer sides of length 920 ft. Write an integral to express the area of the roof of the Pentagon according to these dimensions and evaluate this area.
- 217.** A dodecahedron is a Platonic solid with a surface that consists of 12 pentagons, each of equal area. By how much does the surface area of a dodecahedron increase as the side length of each pentagon doubles from 1 unit to 2 units?
- 218.** An icosahedron is a Platonic solid with a surface that consists of 20 equilateral triangles. By how much does the surface area of an icosahedron increase as the side length of each triangle doubles from a unit to $2a$ units?
- 219.** Write an integral that quantifies the change in the area of the surface of a cube when its side length doubles from s unit to $2s$ units and evaluate the integral.
- 220.** Write an integral that quantifies the increase in the volume of a cube when the side length doubles from s unit to $2s$ units and evaluate the integral.
- 221.** Write an integral that quantifies the increase in the surface area of a sphere as its radius doubles from R unit to $2R$ units and evaluate the integral.
- 222.** Write an integral that quantifies the increase in the volume of a sphere as its radius doubles from R unit to $2R$ units and evaluate the integral.
- 223.** Suppose that a particle moves along a straight line with velocity $v(t) = 4 - 2t$, where $0 \leq t \leq 2$ (in meters per second). Find the displacement at time t and the total distance traveled up to $t = 2$.
- 224.** Suppose that a particle moves along a straight line with velocity defined by $v(t) = t^2 - 3t - 18$, where $0 \leq t \leq 6$ (in meters per second). Find the displacement at time t and the total distance traveled up to $t = 6$.

- 225.** Suppose that a particle moves along a straight line with velocity defined by $v(t) = |2t - 6|$, where $0 \leq t \leq 6$ (in meters per second). Find the displacement at time t and the total distance traveled up to $t = 6$.
- 226.** Suppose that a particle moves along a straight line with acceleration defined by $a(t) = t - 3$, where $0 \leq t \leq 6$ (in meters per second). Find the velocity and displacement at time t and the total distance traveled up to $t = 6$ if $v(0) = 3$ and $d(0) = 0$.
- 227.** A ball is thrown upward from a height of 1.5 m at an initial speed of 40 m/sec. Acceleration resulting from gravity is -9.8 m/sec². Neglecting air resistance, solve for the velocity $v(t)$ and the height $h(t)$ of the ball t seconds after it is thrown and before it returns to the ground.
- 228.** A ball is thrown upward from a height of 3 m at an initial speed of 60 m/sec. Acceleration resulting from gravity is -9.8 m/sec². Neglecting air resistance, solve for the velocity $v(t)$ and the height $h(t)$ of the ball t seconds after it is thrown and before it returns to the ground.
- 229.** The area $A(t)$ of a circular shape is growing at a constant rate. If the area increases from 4π units to 9π units between times $t = 2$ and $t = 3$, find the net change in the radius during that time.
- 230.** A spherical balloon is being inflated at a constant rate. If the volume of the balloon changes from 36π in.³ to 288π in.³ between time $t = 30$ and $t = 60$ seconds, find the net change in the radius of the balloon during that time.

- 231.** Water flows into a conical tank with cross-sectional area πx^2 at height x and volume $\frac{\pi x^3}{3}$ up to height x . If water flows into the tank at a rate of $1 \text{ m}^3/\text{min}$, find the height of water in the tank after 5 min. Find the change in height between 5 min and 10 min.

- 232.** A horizontal cylindrical tank has cross-sectional area $A(x) = 4(6x - x^2) \text{ m}^2$ at height x meters above the bottom when $x \leq 3$.
- The volume V between heights a and b is $\int_a^b A(x) dx$. Find the volume at heights between 2 m and 3 m.
 - Suppose that oil is being pumped into the tank at a rate of 50 L/min . Using the chain rule, $\frac{dx}{dt} = \frac{dx}{dV} \frac{dV}{dt}$, at how many meters per minute is the height of oil in the tank changing, expressed in terms of x , when the height is at x meters?
 - How long does it take to fill the tank to 3 m starting from a fill level of 2 m?

- 233.** The following table lists the electrical power in gigawatts—the rate at which energy is consumed—used in a certain city for different hours of the day, in a typical 24-hour period, with hour 1 corresponding to midnight to 1 a.m.

Hour	Power	Hour	Power
1	28	13	48
2	25	14	49
3	24	15	49
4	23	16	50
5	24	17	50
6	27	18	50
7	29	19	46
8	32	20	43
9	34	21	42
10	39	22	40
11	42	23	37
12	46	24	34

Find the total amount of energy in gigawatt-hours (gW-h) consumed by the city in a typical 24-hour period.

234. The average residential electrical power use (in hundreds of watts) per hour is given in the following table.

Hour	Power	Hour	Power
1	8	13	12
2	6	14	13
3	5	15	14
4	4	16	15
5	5	17	17
6	6	18	19
7	7	19	18
8	8	20	17
9	9	21	16
10	10	22	16
11	10	23	13
12	11	24	11

- Compute the average total energy used in a day in kilowatt-hours (kWh).
- If a ton of coal generates 1842 kWh, how long does it take for an average residence to burn a ton of coal?
- Explain why the data might fit a plot of the form $p(t) = 11.5 - 7.5 \sin\left(\frac{\pi t}{12}\right)$.

235. The data in the following table are used to estimate the average power output produced by Peter Sagan for each of the last 18 sec of Stage 1 of the 2012 Tour de France.

Second	Watts	Second	Watts
1	600	10	1200
2	500	11	1170
3	575	12	1125
4	1050	13	1100
5	925	14	1075
6	950	15	1000
7	1050	16	950
8	950	17	900
9	1100	18	780

Table 5.6 Average Power Output

Source: sportsexercisengineering.com

Estimate the net energy used in kilojoules (kJ), noting that $1W = 1 \text{ J/s}$, and the average power output by Sagan during this time interval.

- 236.** The data in the following table are used to estimate the average power output produced by Peter Sagan for each 15-min interval of Stage 1 of the 2012 Tour de France.

Minutes	Watts	Minutes	Watts
15	200	165	170
30	180	180	220
45	190	195	140
60	230	210	225
75	240	225	170
90	210	240	210
105	210	255	200
120	220	270	220
135	210	285	250
150	150	300	400

Table 5.7 Average Power Output

Source: sportsexercisengineering.com

Estimate the net energy used in kilojoules, noting that $1W = 1 \text{ J/s}$.

- 237.** The distribution of incomes as of 2012 in the United States in \$5000 increments is given in the following table. The k th row denotes the percentage of households with incomes between $5000xk$ and $5000xk + 4999$. The row $k = 40$ contains all households with income between \$200,000 and \$250,000.

0	3.5	11	3.5	21	1.5	31	0.6
1	4.1	12	3.7	22	1.4	32	0.5
2	5.9	13	3.2	23	1.3	33	0.5
3	5.7	14	3.0	24	1.3	34	0.4
4	5.9	15	2.8	25	1.1	35	0.3
5	5.4	16	2.5	26	1.0	36	0.3
6	5.5	17	2.2	27	0.75	37	0.3
7	5.1	18	2.2	28	0.8	38	0.2
8	4.8	19	1.8	29	1.0	39	1.8
9	4.1	20	2.1	30	0.6	40	2.3
10	4.3						

Table 5.8 Income Distributions Source:

<http://www.census.gov/prod/2013pubs/p60-245.pdf>

- Estimate the percentage of U.S. households in 2012 with incomes less than \$55,000.
- What percentage of households had incomes exceeding \$85,000?
- Plot the data and try to fit its shape to that of a graph of the form $a(x + c)e^{-b(x+c)}$ for suitable a, b, c .

- 238.** Newton's law of gravity states that the gravitational force exerted by an object of mass M and one of mass m with centers that are separated by a distance r is $F = G \frac{mM}{r^2}$, with G an empirical constant $G = 6.67 \times 10^{-11} \text{ m}^3 / (\text{kg} \cdot \text{s}^2)$. The work done by a variable force over an interval $[a, b]$ is defined as $W = \int_a^b F(x) dx$. If Earth has mass 5.97219×10^{24} and radius 6371 km, compute the amount of work to elevate a polar weather satellite of mass 1400 kg to its orbiting altitude of 850 km above Earth.
- 239.** For a given motor vehicle, the maximum achievable deceleration from braking is approximately 7 m/sec^2 on dry concrete. On wet asphalt, it is approximately 2.5 m/sec^2 . Given that 1 mph corresponds to 0.447 m/sec , find the total distance that a car travels in meters on dry concrete after the brakes are applied until it comes to a complete stop if the initial velocity is 67 mph (30 m/sec) or if the initial braking velocity is 56 mph (25 m/sec). Find the corresponding distances if the surface is slippery wet asphalt.
- 240.** John is a 25-year old man who weighs 160 lb. He burns $500 - 50t$ calories/hr while riding his bike for t hours. If an oatmeal cookie has 55 cal and John eats cookies at a rate of $4t$ calories during the t th hour, how many net calories has he lost after 3 hours riding his bike?
- 241.** Sandra is a 25-year old woman who weighs 120 lb. She burns $300 - 50t$ cal/hr while walking on her treadmill. Her caloric intake from drinking Gatorade is $100t$ calories/hour during the t th hour. What is her net decrease in calories after walking for 3 hours?
- 242.** A motor vehicle has a maximum efficiency of 33 mpg at a cruising speed of 40 mph. The efficiency drops at a rate of 0.1 mpg/mph between 40 mph and 50 mph, and at a rate of 0.4 mpg/mph between 50 mph and 80 mph. What is the efficiency in miles per gallon if the car is cruising at 50 mph? What is the efficiency in miles per gallon if the car is cruising at 80 mph? If gasoline costs $\$3.50/\text{gal}$, what is the cost of fuel to drive 50 mi at 40 mph, at 50 mph, and at 80 mph?
- 243.** Although some engines are more efficient at given a horsepower than others, on average, fuel efficiency decreases with horsepower at a rate of $1/25$ mpg/horsepower. If a typical 50-horsepower engine has an average fuel efficiency of 32 mpg, what is the average fuel efficiency of an engine with the following horsepower: 150, 300, 450?

244. [T] The following table lists the 2013 schedule of federal income tax versus taxable income.

Taxable Income Range	The Tax Is ...	... Of the Amount Over
\$0–\$8925	10%	\$0
\$8925–\$36,250	\$892.50 + 15%	\$8925
\$36,250–\$87,850	\$4,991.25 + 25%	\$36,250
\$87,850–\$183,250	\$17,891.25 + 28%	\$87,850
\$183,250–\$398,350	\$44,603.25 + 33%	\$183,250
\$398,350–\$400,000	\$115,586.25 + 35%	\$398,350
> \$400,000	\$116,163.75 + 39.6%	\$400,000

Table 5.9 Federal Income Tax Versus Taxable Income Source: <http://www.irs.gov/pub/irs-prior/i1040tt--2013.pdf>.

Suppose that Steve just received a \$10,000 raise. How much of this raise is left after federal taxes if Steve's salary before receiving the raise was \$40,000? If it was \$90,000? If it was \$385,000?

245. [T] The following table provides hypothetical data regarding the level of service for a certain highway.

Highway Speed Range (mph)	Vehicles per Hour per Lane	Density Range (vehicles/mi)
> 60	< 600	< 10
60–57	600–1000	10–20
57–54	1000–1500	20–30
54–46	1500–1900	30–45
46–30	1900–2100	45–70
<30	Unstable	70–200

Table 5.10

- Plot vehicles per hour per lane on the x-axis and highway speed on the y-axis.
- Compute the average decrease in speed (in miles per hour) per unit increase in congestion (vehicles per hour per lane) as the latter increases from 600 to 1000, from 1000 to 1500, and from 1500 to 2100. Does the decrease in miles per hour depend linearly on the increase in vehicles per hour per lane?
- Plot minutes per mile (60 times the reciprocal of miles per hour) as a function of vehicles per hour per lane. Is this function linear?

For the next two exercises use the data in the following table, which displays bald eagle populations from 1963 to 2000 in the continental United States.

Year	Population of Breeding Pairs of Bald Eagles
1963	487
1974	791
1981	1188

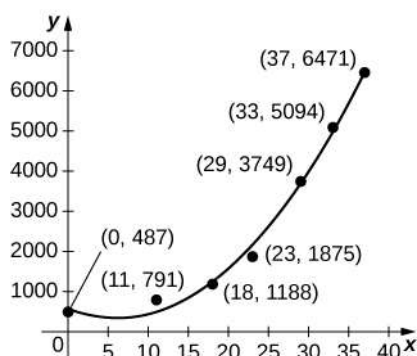
Table 5.11 Population of Breeding Bald Eagle Pairs Source: <http://www.fws.gov/Midwest/eagle/population/chtotfprs.html>.

Year	Population of Breeding Pairs of Bald Eagles
1986	1875
1992	3749
1996	5094
2000	6471

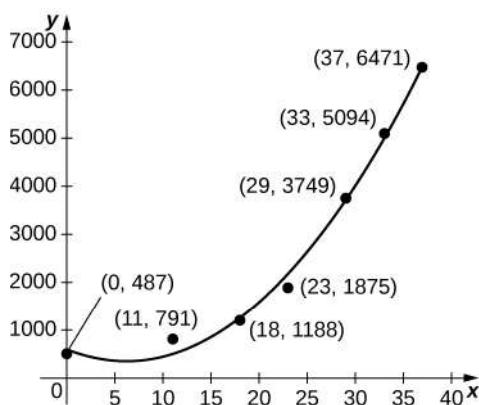
Table 5.11 Population of Breeding Bald Eagle Pairs

Source: <http://www.fws.gov/Midwest/eagle/population/ctofprs.html>.

- 246. [T]** The graph below plots the quadratic $p(t) = 6.48t^2 - 80.31t + 585.69$ against the data in preceding table, normalized so that $t = 0$ corresponds to 1963. Estimate the average number of bald eagles per year present for the 37 years by computing the average value of p over $[0, 37]$.

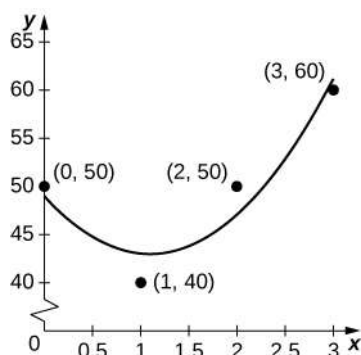


- 247. [T]** The graph below plots the cubic $p(t) = 0.07t^3 + 2.42t^2 - 25.63t + 521.23$ against the data in the preceding table, normalized so that $t = 0$ corresponds to 1963. Estimate the average number of bald eagles per year present for the 37 years by computing the average value of p over $[0, 37]$.



248. [T] Suppose you go on a road trip and record your speed at every half hour, as compiled in the following table. The best quadratic fit to the data is $q(t) = 5x^2 - 11x + 49$, shown in the accompanying graph. Integrate q to estimate the total distance driven over the 3 hours.

Time (hr)	Speed (mph)
0 (start)	50
1	40
2	50
3	60



As a car accelerates, it does not accelerate at a constant rate; rather, the acceleration is variable. For the following exercises, use the following table, which contains the acceleration measured at every second as a driver merges onto a freeway.

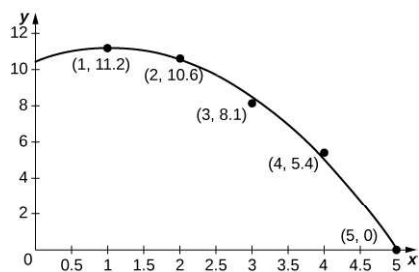
Time (sec)	Acceleration (mph/sec)
1	11.2
2	10.6
3	8.1
4	5.4
5	0

As a car accelerates, it does not accelerate at a constant rate; rather, the acceleration is variable. For the next three exercises use the following table, which contains the acceleration measured at every second as a driver merges onto a freeway.

Time (sec)	Acceleration (mph/sec)
1	11.2
2	10.6
3	8.1
4	5.4
5	0

Table 5.12

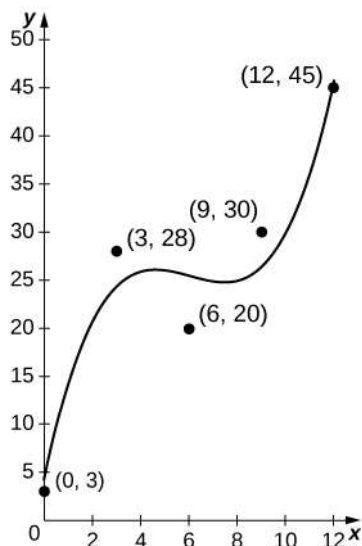
249. [T] The accompanying graph plots the best quadratic fit, $a(t) = -0.70t^2 + 1.44t + 10.44$, to the data from the preceding table. Compute the average value of $a(t)$ to estimate the average acceleration between $t = 0$ and $t = 5$.



250. [T] Using your acceleration equation from the previous exercise, find the corresponding velocity equation. Assuming the initial velocity is 65 mph, find the velocity at time $t = 0$.
251. [T] Using your velocity equation from the previous exercise, find the corresponding distance equation, assuming your initial distance is 0 mi. How far did you travel while you accelerated your car? (Hint: You will need to convert time units.)

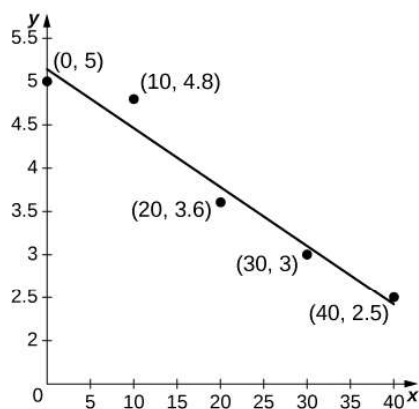
- 252. [T]** The number of hamburgers sold at a restaurant throughout the day is given in the following table, with the accompanying graph plotting the best cubic fit to the data, $b(t) = 0.12t^3 - 2.13t^2 + 12.13t + 3.91$, with $t = 0$ corresponding to 9 a.m. and $t = 12$ corresponding to 9 p.m. Compute the average value of $b(t)$ to estimate the average number of hamburgers sold per hour.

Hours Past Midnight	No. of Burgers Sold
9	3
12	28
15	20
18	30
21	45



- 253. [T]** An athlete runs by a motion detector, which records her speed, as displayed in the following table. The best linear fit to this data, $\ell(t) = -0.068t + 5.14$, is shown in the accompanying graph. Use the average value of $\ell(t)$ between $t = 0$ and $t = 40$ to estimate the runner's average speed.

Minutes	Speed (m/sec)
0	5
10	4.8
20	3.6
30	3.0
40	2.5



5.5 Substitution

Learning Objectives

- 5.5.1** Use substitution to evaluate indefinite integrals.
5.5.2 Use substitution to evaluate definite integrals.

The Fundamental Theorem of Calculus gave us a method to evaluate integrals without using Riemann sums. The drawback of this method, though, is that we must be able to find an antiderivative, and this is not always easy. In this section we examine a technique, called **integration by substitution**, to help us find antiderivatives. Specifically, this method helps us find antiderivatives when the integrand is the result of a chain-rule derivative.

At first, the approach to the substitution procedure may not appear very obvious. However, it is primarily a visual task—that is, the integrand shows you what to do; it is a matter of recognizing the form of the function. So, what are we supposed to see? We are looking for an integrand of the form $f[g(x)]g'(x)dx$. For example, in the integral

$\int (x^2 - 3)^3 2x dx$, we have $f(x) = x^3$, $g(x) = x^2 - 3$, and $g'(x) = 2x$. Then,

$$f[g(x)]g'(x) = (x^2 - 3)^3 (2x),$$

and we see that our integrand is in the correct form.

The method is called *substitution* because we substitute part of the integrand with the variable u and part of the integrand with du . It is also referred to as **change of variables** because we are changing variables to obtain an expression that is easier to work with for applying the integration rules.

Theorem 5.7

Substitution with Indefinite Integrals

Let $u = g(x)$, where $g'(x)$ is continuous over an interval, let $f(x)$ be continuous over the corresponding range of g , and let $F(x)$ be an antiderivative of $f(x)$. Then,

$$\begin{aligned} \int f[g(x)]g'(x)dx &= \int f(u)du \\ &= F(u) + C \\ &= F(g(x)) + C. \end{aligned} \tag{5.19}$$

Proof

Let f , g , u , and F be as specified in the theorem. Then

$$\begin{aligned} \frac{d}{dx} F(g(x)) &= F'(g(x))g'(x) \\ &= f[g(x)]g'(x). \end{aligned}$$

Integrating both sides with respect to x , we see that

$$\int f[g(x)]g'(x)dx = F(g(x)) + C.$$

If we now substitute $u = g(x)$, and $du = g'(x)dx$, we get

$$\begin{aligned} \int f[g(x)]g'(x)dx &= \int f(u)du \\ &= F(u) + C \\ &= F(g(x)) + C. \end{aligned}$$

□

Returning to the problem we looked at originally, we let $u = x^2 - 3$ and then $du = 2x dx$. Rewrite the integral in terms of u :

$$\int \underbrace{(x^2 - 3)^3}_u \underbrace{(2x dx)}_{du} = \int u^3 du.$$

Using the power rule for integrals, we have

$$\int u^3 du = \frac{u^4}{4} + C.$$

Substitute the original expression for x back into the solution:

$$\frac{u^4}{4} + C = \frac{(x^2 - 3)^4}{4} + C.$$

We can generalize the procedure in the following Problem-Solving Strategy.

Problem-Solving Strategy**Integration by Substitution**

1. Look carefully at the integrand and select an expression $g(x)$ within the integrand to set equal to u . Let's select $g(x)$ such that $g'(x)$ is also part of the integrand.
2. Substitute $u = g(x)$ and $du = g'(x)dx$ into the integral.
3. We should now be able to evaluate the integral with respect to u . If the integral can't be evaluated we need to go back and select a different expression to use as u .
4. Evaluate the integral in terms of u .
5. Write the result in terms of x and the expression $g(x)$.

EXAMPLE 5.30**Using Substitution to Find an Antiderivative**

Use substitution to find the antiderivative $\int 6x(3x^2 + 4)^4 dx$.

 Solution

The first step is to choose an expression for u . We choose $u = 3x^2 + 4$ because then $du = 6x dx$, and we already have du in the integrand. Write the integral in terms of u :

$$\int 6x(3x^2 + 4)^4 dx = \int u^4 du.$$

Remember that du is the derivative of the expression chosen for u , regardless of what is inside the integrand. Now we can evaluate the integral with respect to u :

$$\begin{aligned} \int u^4 du &= \frac{u^5}{5} + C \\ &= \frac{(3x^2 + 4)^5}{5} + C. \end{aligned}$$

 Analysis

We can check our answer by taking the derivative of the result of integration. We should obtain the integrand. Picking a value for C of 1, we let $y = \frac{1}{5}(3x^2 + 4)^5 + 1$. We have

$$y = \frac{1}{5}(3x^2 + 4)^5 + 1,$$

so

$$\begin{aligned} y' &= \left(\frac{1}{5}\right) 5(3x^2 + 4)^4 6x \\ &= 6x(3x^2 + 4)^4. \end{aligned}$$

This is exactly the expression we started with inside the integrand.

 5.25 Use substitution to find the antiderivative $\int 3x^2(x^3 - 3)^2 dx$.

Sometimes we need to adjust the constants in our integral if they don't match up exactly with the expressions we are substituting.

EXAMPLE 5.31**Using Substitution with Alteration**

Use substitution to find $\int z\sqrt{z^2 - 5} dz$.

✓ Solution

Rewrite the integral as $\int z(z^2 - 5)^{1/2} dz$. Let $u = z^2 - 5$ and $du = 2z dz$. Now we have a problem because $du = 2z dz$ and the original expression has only $z dz$. We have to alter our expression for du or the integral in u will be twice as large as it should be. If we multiply both sides of the du equation by $\frac{1}{2}$, we can solve this problem. Thus,

$$\begin{aligned} u &= z^2 - 5 \\ du &= 2z dz \\ \frac{1}{2} du &= \frac{1}{2}(2z) dz = z dz. \end{aligned}$$

Write the integral in terms of u , but pull the $\frac{1}{2}$ outside the integration symbol:

$$\int z(z^2 - 5)^{1/2} dz = \frac{1}{2} \int u^{1/2} du.$$

Integrate the expression in u :

$$\begin{aligned} \frac{1}{2} \int u^{1/2} du &= \left(\frac{1}{2}\right) \frac{u^{3/2}}{\frac{3}{2}} + C \\ &= \left(\frac{1}{2}\right) \left(\frac{2}{3}\right) u^{3/2} + C \\ &= \frac{1}{3} u^{3/2} + C \\ &= \frac{1}{3} (z^2 - 5)^{3/2} + C. \end{aligned}$$

✓ 5.26 Use substitution to find $\int x^2(x^3 + 5)^9 dx$.

EXAMPLE 5.32**Using Substitution with Integrals of Trigonometric Functions**

Use substitution to evaluate the integral $\int \frac{\sin t}{\cos^3 t} dt$.

✓ Solution

We know the derivative of $\cos t$ is $-\sin t$, so we set $u = \cos t$. Then $du = -\sin t dt$. Substituting into the integral, we have

$$\int \frac{\sin t}{\cos^3 t} dt = - \int \frac{du}{u^3}.$$

Evaluating the integral, we get

$$\begin{aligned} - \int \frac{du}{u^3} &= - \int u^{-3} du \\ &= - \left(-\frac{1}{2}\right) u^{-2} + C. \end{aligned}$$

Putting the answer back in terms of t , we get

$$\begin{aligned}\int \frac{\sin t}{\cos^3 t} dt &= \frac{1}{2u^2} + C \\ &= \frac{1}{2\cos^2 t} + C.\end{aligned}$$

✓ 5.27 Use substitution to evaluate the integral $\int \frac{\cos t}{\sin^2 t} dt$.

Sometimes we need to manipulate an integral in ways that are more complicated than just multiplying or dividing by a constant. We need to eliminate all the expressions within the integrand that are in terms of the original variable. When we are done, u should be the only variable in the integrand. In some cases, this means solving for the original variable in terms of u . This technique should become clear in the next example.

EXAMPLE 5.33

Finding an Antiderivative Using u -Substitution

Use substitution to find the antiderivative $\int \frac{x}{\sqrt{x-1}} dx$.

✓ Solution

If we let $u = x - 1$, then $du = dx$. But this does not account for the x in the numerator of the integrand. We need to express x in terms of u . If $u = x - 1$, then $x = u + 1$. Now we can rewrite the integral in terms of u :

$$\begin{aligned}\int \frac{x}{\sqrt{x-1}} dx &= \int \frac{u+1}{\sqrt{u}} du \\ &= \int \sqrt{u} + \frac{1}{\sqrt{u}} du \\ &= \int (u^{1/2} + u^{-1/2}) du.\end{aligned}$$

Then we integrate in the usual way, replace u with the original expression, and factor and simplify the result. Thus,

$$\begin{aligned}\int (u^{1/2} + u^{-1/2}) du &= \frac{2}{3}u^{3/2} + 2u^{1/2} + C \\ &= \frac{2}{3}(x-1)^{3/2} + 2(x-1)^{1/2} + C \\ &= (x-1)^{1/2} \left[\frac{2}{3}(x-1) + 2 \right] + C \\ &= (x-1)^{1/2} \left(\frac{2}{3}x - \frac{2}{3} + \frac{6}{3} \right) + C \\ &= (x-1)^{1/2} \left(\frac{2}{3}x + \frac{4}{3} \right) + C \\ &= \frac{2}{3}(x-1)^{1/2} (x+2) + C.\end{aligned}$$

✓ 5.28 Use substitution to evaluate the indefinite integral $\int \cos^3 t \sin t dt$.

Substitution for Definite Integrals

Substitution can be used with definite integrals, too. However, using substitution to evaluate a definite integral requires a change to the limits of integration. If we change variables in the integrand, the limits of integration change as well.

Theorem 5.8**Substitution with Definite Integrals**

Let $u = g(x)$ and let g' be continuous over an interval $[a, b]$, and let f be continuous over the range of $u = g(x)$. Then,

$$\int_a^b f(g(x)) g'(x) dx = \int_{g(a)}^{g(b)} f(u) du.$$

Although we will not formally prove this theorem, we justify it with some calculations here. From the substitution rule for indefinite integrals, if $F(x)$ is an antiderivative of $f(x)$, we have

$$\int f(g(x)) g'(x) dx = F(g(x)) + C.$$

Then

$$\begin{aligned} \int_a^b f[g(x)] g'(x) dx &= F(g(x)) \Big|_{x=a}^{x=b} \\ &= F(g(b)) - F(g(a)) \\ &= F(u) \Big|_{u=g(a)}^{u=g(b)} \\ &= \int_{g(a)}^{g(b)} f(u) du, \end{aligned} \tag{5.20}$$

and we have the desired result.

EXAMPLE 5.34**Using Substitution to Evaluate a Definite Integral**

Use substitution to evaluate $\int_0^1 x^2 (1 + 2x^3)^5 dx$.

 Solution

Let $u = 1 + 2x^3$, so $du = 6x^2 dx$. Since the original function includes one factor of x^2 and $du = 6x^2 dx$, multiply both sides of the du equation by $1/6$. Then,

$$\begin{aligned} du &= 6x^2 dx \\ \frac{1}{6} du &= x^2 dx. \end{aligned}$$

To adjust the limits of integration, note that when $x = 0$, $u = 1 + 2(0) = 1$, and when $x = 1$, $u = 1 + 2(1) = 3$. Then

$$\int_0^1 x^2 (1 + 2x^3)^5 dx = \frac{1}{6} \int_1^3 u^5 du.$$

Evaluating this expression, we get

$$\begin{aligned} \frac{1}{6} \int_1^3 u^5 du &= \left(\frac{1}{6} \right) \left(\frac{u^6}{6} \right) \Big|_1^3 \\ &= \frac{1}{36} [(3)^6 - (1)^6] \\ &= \frac{182}{9}. \end{aligned}$$

- ✓ 5.29 Use substitution to evaluate the definite integral $\int_{-1}^0 y(2y^2 - 3)^5 dy$.

EXAMPLE 5.35**Using Substitution with an Exponential Function**

Use substitution to evaluate $\int_0^1 xe^{4x^2+3} dx$.

✓ Solution

Let $u = 4x^2 + 3$. Then, $du = 8xdx$. To adjust the limits of integration, we note that when $x = 0$, $u = 3$, and when $x = 1$, $u = 7$. So our substitution gives

$$\begin{aligned}\int_0^1 xe^{4x^2+3} dx &= \frac{1}{8} \int_3^7 e^u du \\ &= \frac{1}{8} e^u \Big|_3^7 \\ &= \frac{e^7 - e^3}{8} \\ &\approx 134.568.\end{aligned}$$

- ✓ 5.30 Use substitution to evaluate $\int_0^1 x^2 \cos\left(\frac{\pi}{2}x^3\right) dx$.

Substitution may be only one of the techniques needed to evaluate a definite integral. All of the properties and rules of integration apply independently, and trigonometric functions may need to be rewritten using a trigonometric identity before we can apply substitution. Also, we have the option of replacing the original expression for u after we find the antiderivative, which means that we do not have to change the limits of integration. These two approaches are shown in [Example 5.36](#).

EXAMPLE 5.36**Using Substitution to Evaluate a Trigonometric Integral**

Use substitution to evaluate $\int_0^{\pi/2} \cos^2 \theta d\theta$.

✓ Solution

Let us first use a trigonometric identity to rewrite the integral. The trig identity $\cos^2 \theta = \frac{1+\cos 2\theta}{2}$ allows us to rewrite the integral as

$$\int_0^{\pi/2} \cos^2 \theta d\theta = \int_0^{\pi/2} \frac{1 + \cos 2\theta}{2} d\theta.$$

Then,

$$\begin{aligned}\int_0^{\pi/2} \left(\frac{1 + \cos 2\theta}{2}\right) d\theta &= \int_0^{\pi/2} \left(\frac{1}{2} + \frac{1}{2}\cos 2\theta\right) d\theta \\ &= \frac{1}{2} \int_0^{\pi/2} d\theta + \frac{1}{2} \int_0^{\pi/2} \cos 2\theta d\theta.\end{aligned}$$

We can evaluate the first integral as it is, but we need to make a substitution to evaluate the second integral. Let $u = 2\theta$. Then, $du = 2d\theta$, or $\frac{1}{2}du = d\theta$. Also, when $\theta = 0$, $u = 0$, and when $\theta = \pi/2$, $u = \pi$. Expressing the second integral in terms

of u , we have

$$\begin{aligned}\frac{1}{2} \int_0^{\pi/2} d\theta + \frac{1}{2} \int_0^{\pi/2} \cos 2\theta d\theta &= \frac{1}{2} \int_0^{\pi/2} d\theta + \frac{1}{2} \left(\frac{1}{2} \right) \int_0^{\pi} \cos u du \\ &= \frac{\theta}{2} \Big|_{\theta=0}^{\theta=\pi/2} + \frac{1}{4} \sin u \Big|_{u=0}^{u=\pi} \\ &= \left(\frac{\pi}{4} - 0 \right) + (0 - 0) = \frac{\pi}{4}.\end{aligned}$$



SECTION 5.5 EXERCISES

- 254.** Why is u -substitution referred to as *change of variable*?
- 255.** 2. If $f = g \circ h$, when reversing the chain rule, $\frac{d}{dx}(g \circ h)(x) = g'(h(x))h'(x)$, should you take $u = g(x)$ or $u = h(x)$?

In the following exercises, verify each identity using differentiation. Then, using the indicated u -substitution, identify f such that the integral takes the form $\int f(u) du$.

- 256.** $\int x\sqrt{x+1} dx = \frac{2}{15}(x+1)^{3/2}(3x-2) + C; u = x+1$
- 257.** For $x > 1$: $\int \frac{x^2}{\sqrt{x-1}} dx = \frac{2}{15}\sqrt{x-1}(3x^2+4x+8) + C; u = x-1$
- 258.** $\int x\sqrt{4x^2+9} dx = \frac{1}{12}(4x^2+9)^{3/2} + C; u = 4x^2+9$
- 259.** $\int \frac{x}{\sqrt{4x^2+9}} dx = \frac{1}{4}\sqrt{4x^2+9} + C; u = 4x^2+9$ **260.** $\int \frac{x}{(4x^2+9)^2} dx = -\frac{1}{8(4x^2+9)}; u = 4x^2+9$

In the following exercises, find the antiderivative using the indicated substitution.

- 261.** $\int (x+1)^4 dx; u = x+1$ **262.** $\int (x-1)^5 dx; u = x-1$ **263.** $\int (2x-3)^{-7} dx; u = 2x-3$
- 264.** $\int (3x-2)^{-11} dx; u = 3x-2$ **265.** $\int \frac{x}{\sqrt{x^2+1}} dx; u = x^2+1$ **266.** $\int \frac{x}{\sqrt{1-x^2}} dx; u = 1-x^2$
- 267.** $\int (x-1)(x^2-2x)^3 dx; u = x^2-2x$ **268.** $\int (x^2-2x)(x^3-3x^2)^2 dx; u = x^3-3x^2$
- 269.** $\int \cos^3 \theta d\theta; u = \sin \theta$ **270.** $\int \sin^3 \theta d\theta; u = \cos \theta$
(Hint: $\cos^2 \theta = 1 - \sin^2 \theta$) (Hint: $\sin^2 \theta = 1 - \cos^2 \theta$)

In the following exercises, use a suitable change of variables to determine the indefinite integral.

- 271.** $\int x(1-x)^{99} dx$ **272.** $\int t(1-t^2)^{10} dt$ **273.** $\int (11x-7)^{-3} dx$
- 274.** $\int (7x-11)^4 dx$ **275.** $\int \cos^3 \theta \sin \theta d\theta$ **276.** $\int \sin^7 \theta \cos \theta d\theta$

277. $\int \cos^2(\pi t) \sin(\pi t) dt$ 278. $\int \sin^2 x \cos^3 x dx$ 279. $\int t \sin(t^2) \cos(t^2) dt$
 (Hint: $\sin^2 x + \cos^2 x = 1$)
280. $\int t^2 \cos^2(t^3) \sin(t^3) dt$ 281. $\int \frac{x^2}{(x^3 - 3)^2} dx$ 282. $\int \frac{x^3}{\sqrt{1 - x^2}} dx$
283. $\int \frac{y^5}{(1 - y^3)^{3/2}} dy$ 284. $\int \cos \theta (1 - \cos \theta)^{99} \sin \theta d\theta$ 285. $\int (1 - \cos^3 \theta)^{10} \cos^2 \theta \sin \theta d\theta$
286. $\int (\cos \theta - 1)(\cos^2 \theta - 2 \cos \theta)^3 \sin \theta d\theta$ 287. $\int (\sin^2 \theta - 2 \sin \theta)(\sin^3 \theta - 3 \sin^2 \theta)^3 \cos \theta d\theta$

In the following exercises, use a calculator to estimate the area under the curve using left Riemann sums with 50 terms, then use substitution to solve for the exact answer.

288. [T] $y = 3(1 - x)^2$ over $[0, 2]$ 289. [T] $y = x(1 - x^2)^3$ over $[-1, 2]$ 290. [T] $y = \sin x(1 - \cos x)^2$ over $[0, \pi]$
291. [T] $y = \frac{x}{(x^2 + 1)^2}$ over $[-1, 1]$

In the following exercises, use a change of variables to evaluate the definite integral.

292. $\int_0^1 x \sqrt{1 - x^2} dx$ 293. $\int_0^1 \frac{x}{\sqrt{1 + x^2}} dx$ 294. $\int_0^2 \frac{t}{\sqrt{5 + t^2}} dt$
295. $\int_0^1 \frac{t^2}{\sqrt{1 + t^3}} dt$ 296. $\int_0^{\pi/4} \sec^2 \theta \tan \theta d\theta$ 297. $\int_0^{\pi/4} \frac{\sin \theta}{\cos^4 \theta} d\theta$

In the following exercises, evaluate the indefinite integral $\int f(x) dx$ with constant $C = 0$ using u -substitution. Then, graph the function and the antiderivative over the indicated interval. If possible, estimate a value of C that would need to be added to the antiderivative to make it equal to the definite integral $F(x) = \int_a^x f(t) dt$, with a the left endpoint of the given interval.

298. [T] $\int (2x + 1)e^{x^2 + x - 6} dx$ over $[-3, 2]$ 299. [T] $\int \frac{\cos(\ln(2x))}{x} dx$ on $[0, 2]$ 300. [T] $\int \frac{3x^2 + 2x + 1}{\sqrt{x^3 + x^2 + x + 4}} dx$ over $[-1, 2]$
301. [T] $\int \frac{\sin x}{\cos^3 x} dx$ over $[-\frac{\pi}{3}, \frac{\pi}{3}]$ 302. [T] $\int (x + 2)e^{-x^2 - 4x + 3} dx$ over $[-5, 1]$ 303. [T] $\int 3x^2 \sqrt{2x^3 + 1} dx$ over $[0, 1]$
304. If $h(a) = h(b)$ in $\int_a^b g'(h(x)) h'(x) dx$, what can you say about the value of the integral?
305. Is the substitution $u = 1 - x^2$ in the definite integral $\int_0^2 \frac{x}{1 - x^2} dx$ okay? If not, why not?

In the following exercises, use a change of variables to show that each definite integral is equal to zero.

306. $\int_0^\pi \cos^2(2\theta) \sin(2\theta) d\theta$

307. $\int_0^{\sqrt{\pi}} t \cos(t^2) \sin(t^2) dt$

308. $\int_0^1 (1-2t) dt$

309. $\int_0^1 \frac{1-2t}{\left(1+\left(t-\frac{1}{2}\right)^2\right)} dt$

310. $\int_0^\pi \sin\left(\left(t-\frac{\pi}{2}\right)^3\right) \cos\left(t-\frac{\pi}{2}\right) dt$

311. $\int_0^2 (1-t) \cos(\pi t) dt$

312. $\int_{\pi/4}^{3\pi/4} \sin^2 t \cos t dt$

313. Show that the average value of $f(x)$ over an interval $[a, b]$ is the same as the average value of $f(cx)$ over the interval $\left[\frac{a}{c}, \frac{b}{c}\right]$ for $c > 0$.

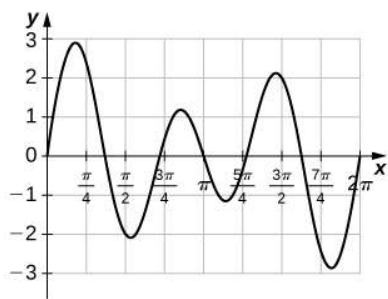
314. Find the area under the graph of $f(t) = \frac{t}{(1+t^2)^a}$ between $t = 0$ and $t = x$ where $a > 0$ and $a \neq 1$ is fixed, and evaluate the limit as $x \rightarrow \infty$.

315. Find the area under the graph of $g(t) = \frac{t}{(1-t^2)^a}$ between $t = 0$ and $t = x$, where $0 < x < 1$ and $a > 0$ is fixed. Evaluate the limit as $x \rightarrow 1$.

316. The area of a semicircle of radius 1 can be expressed as $\int_{-1}^1 \sqrt{1-x^2} dx$. Use the substitution $x = \cos t$ to express the area of a semicircle as the integral of a trigonometric function. You do not need to compute the integral.

317. The area of the top half of an ellipse with a major axis that is the x -axis from $x = -a$ to $x = a$ and with a minor axis that is the y -axis from $y = -b$ to $y = b$ can be written as $\int_{-a}^a b \sqrt{1 - \frac{x^2}{a^2}} dx$. Use the substitution $x = a \cos t$ to express this area in terms of an integral of a trigonometric function. You do not need to compute the integral.

318. [T] The following graph is of a function of the form $f(t) = a \sin(nt) + b \sin(mt)$. Estimate the coefficients a and b , and the frequency parameters n and m . Use these estimates to approximate $\int_0^\pi f(t) dt$.



319. [T] The following graph is of a function of the form $f(x) = a \cos(nx) + b \cos(mx)$. Estimate the coefficients a and b and the frequency parameters n and m . Use these estimates to approximate $\int_0^{\pi/2} f(t) dt$.

